SAKSHYA

साक्ष्य / Evidence

AICTE security deposits, with evidence at every step.

A single, accountable journey from institute FDR submission to verified release and settlement.

SIH 2026

PROTOTYPE v1.0

Evidence-led workflow / Role-based control / Verifiable letters

Sakshya = evidence / witness | Source: SRS-SAK-001, §§1.1–1.4

A deposit is only as trustworthy as its evidence.

The workflow breaks when records, verification and deadlines live in different places.

Scattered records

Files, email and registers create fragmented views of institute obligations and FDRs.

Claims mistaken for facts

An uploaded receipt is an institute claim; it is not independent proof of a valid lien.

Time depends on memory

Maturity, renewal and follow-up dates can be missed when tracking is manual.

Release lacks a visible chain

Eligibility, signing, bank execution and AICTE credit need distinct owners and evidence.

Design response: one versioned case record, explicit trust levels and auditable state changes.

Source: SRS §§1.2–1.3

One case record. Evidence grows; authority stays separated.

Sakshya connects institute, bank, officer, Release Authority and Finance around a shared lifecycle.

01

Submit

Institute submits FDR details and document.

>

02

Verify

Checks raise explainable flags; bank confirms directly.

>

03

Monitor

Track obligation, maturity, renewal and exceptions.

>

04

Release

Eligibility, signed letter, bank execution, settlement.

No single upload or automated flag can promote a claimed value to a bank-confirmed fact.

Source: SRS §§1.2, 7.2, 9.1

Trust is field-level—not a green tick on a whole file.

Evidence, provenance and uncertainty remain visible as a case moves forward.

L0

Claimed

Institute-submitted value
Evidence: submitted document

L1

Attested

Officer-reviewed value
Evidence: review and audit

L2

Bank-confirmed

Bank-confirmed value
Evidence: direct bank response

U1 Trust chain

U2 Hybrid lien: institute authorises; bank confirms directly to AICTE.

U4 Document checks flag inconsistencies; they never approve or reject a deposit.

Source: SRS §§1.4, 1.5; U1–U4

Designed for explainability, continuity and accountability.

Five controls turn the workflow into a chain of evidence—not just a form and a status.

U3 Verifiable release letter

Hash-bound, QR-linked, time-limited and revocable; the bank checks status before acting.

U5 Continuous assurance

Risk-based bank reconciliation helps surface lien changes and other exceptions.

U6 Rules as configuration

Policy values are versioned/configurable rather than buried in business logic.

U7 Explainable priority

Officers see why a case is prioritised, not only a score.

U8 Tamper-evident audit / State changes are attributable and linked in an integrity-checkable trail.

Source: SRS §1.4; U3, U5–U8

A simple prototype stack; production adapters stay explicit.

The v1.0 plan prioritises a runnable vertical slice and honest simulator boundaries.

PEOPLE & WORKSPACES

Institute • Officer • Bank • Release Authority • Finance • Public verifier

WEB APP

React + TypeScript / Vite / role-aware routes / English + Hindi

API & DOMAIN

FastAPI modules: intake • bank • monitoring • release • insights

Role checks / state machines / policy / audit chain / API contracts

Prototype: SQLite / local files / in-process jobs

Adapters: portal bank / simulated signer / local IdP

Source: 4-Day Plan §§1.2–1.3; SRS §§3.1–3.6, Appendix H

From an institute claim to evidence-backed activation.

Planned walkthrough: submission, explainable flags, direct bank confirmation and role separation.

01 INSTITUTE

Submit

Upload genuine FDR
Review extracted fields
Submit as L0 claim

02 CHECKS

Inspect

Show plain-language flags
Compare document fields
Flags are not decisions

03 BANK

Confirm

Use verified directory
Maker + checker review
Critical fields move to L2

04 OFFICER

Activate

Show SoD rejection
Use a different approver
Activate after checks

Demo truth: seed data and backend availability determine which transitions can be shown live.

Source: 4-Day Plan §1.1 G1–G8; SRS E2E-02, 03, 08, 09

A release is not complete when a letter is signed.

Show the independent checks from eligibility to final AICTE-share settlement.

Eligibility

Hold blocks release
Show reason + fix hint
Lift hold; AO recommends

Signed letter

RA completes step-up
PDF carries QR + hash
14-day validity per spec

Bank execution

Public status check
Maker / checker record UTR
Replay is rejected

Finance + audit

Confirm AICTE share
Mismatch needs a reason
Close; verify audit chain

A signed release is verifiable, single-use and revocable by design (per SRS); do not claim a live bank transfer.

Source: 4-Day Plan §1.1 G9–G12; SRS E2E-13…17

Measure outcomes; do not invent a baseline.

The project defines what to measure. No before/after improvement is claimed in this deck.

Record accuracy	Instruments with L2 on every critical field / eligible instruments	MEASURED FROM DATA
Verification turnaround	Submit → bank-confirmed median and p90	MEASURED FROM TIMESTAMPS
Timely renewals	Renewals accepted before maturity / renewals due	MEASURED FROM CASES
Manual effort	Automated actions counted; officer touch time estimated from audit events	ESTIMATE; BASELINE SOURCE REQUIRED
Fraud-catch	Precision / recall on labelled synthetic corpus only	SYNTHETIC; NOT FIELD ACCURACY

Source: SRS T-202; §9.1

Ten expected capabilities, mapped to visible evidence.

Use this as a review checklist; feature availability and live API state vary by module.

01 Repository

Institute obligations + FDR register

02 Documents

FDR upload, preview + records

03 Verification

Bank confirmation + discrepancy flags

04 Validity

Maturity windows + alerts

05 Renewal

Renewal and replacement tracking

06 Release

Eligibility → sign → verify → settle

07 Alerts

Notifications + escalation

08 Reporting

KPI dashboard + reports

09 Audit & access

Role guards + integrity-checkable audit

10 Integration

Bank/signer adapters + public verifier

Source: SRS §9.1; capability names condensed

Checklist status is not a certification: verify each item against the current build and its test evidence.

What v1.0 demonstrates—and what remains a boundary.

A prototype can prove workflow logic without pretending to be a production integration.

Demonstrates in the prototype

Role-based journeys and interfaces
Rule-driven workflow states
Synthetic, fictional demo records
API contracts and adapter seams

Not a production integration

No real bank core/SFMS connection
No fund movement by Sakshya
Software signing is simulated
Policy defaults still need validation

Safe interpretation

Document heuristics surface flags
A flag is not proof of fraud
Bank confirmation is a distinct step
Human authority remains explicit

Current verification status

PA-1–PA-4 frontend flows exist
Build and lint checks pass
Selected flows tested with mocks
Live end-to-end depends on API readiness

Data labels: synthetic = demo corpus / assumed = unvalidated baseline / measured = computed from a stated dataset.

Source: 4-Day Plan §§1.2–1.3, 10; SRS Appendix H

Move from a runnable slice to an operable public service.

Sequence the next investments around verified needs, integrations and governance.

NOW / PROTOTYPE

Prove the golden path

Complete backend/API integration
Rehearse seeded scenarios
Close known issues + test evidence

NEXT / HARDEN

Strengthen operations

PostgreSQL + migrations
Object storage + job queue
CI, security and observability

LATER / CONNECT

Integrate with institutions

Bank channel adapters
Provider-backed eSign + KMS/HSM
Government hosting + validated policy

Also validate regulatory/policy values with AICTE and mentors before operational use.

Source: 4-Day Plan §10; SRS Appendix H

Hard questions deserve precise answers.

Use the boundary—not a promise—as the answer when evidence is not yet available.

Bank API availability?

Not assumed. The portal channel is the initial path; API/bulk adapters need bank-by-bank onboarding.

What is the legal basis for lien?

The prototype does not adjudicate legal authority. AICTE must validate the mandate, rules and prescribed format.

Can forgery detection prove fraud?

No. OCR/QR/consistency checks raise explainable flags; independent bank confirmation is required.

Will it scale?

The prototype stack is deliberately small. Production needs migration, storage, queue, security and performance work.

How is privacy protected?

Use minimum necessary data, role/scope controls and audit; demo data is synthetic. Production governance remains required.

Can it run in government hosting?

Designed without mandatory SaaS; deployment, data residency, secrets and KMS/HSM need production hardening.

Source: SRS §§1.2, 2.5, 3.6, Appendix H; 4-Day Plan §§1.2, 10

A 7-minute planned route through the six test scenarios.

Planned timings only—not measured demo duration. Use reset seed state; keep a recording as fallback.

0:00–1:20	**New FDR submission**	G1–G2 / S1; show claimed fields and review.
1:20–2:20	**Discrepancy + verification**	G3–G5, G8 / S2, S6; show flags, bank check, SoD.
2:20–3:15	**Validity + renewal**	G7 / S3–S5; advance demo clock, alert, renew.
3:15–4:30	**Release eligibility**	G9 / S7; show hold, remedy, recommendation.
4:30–5:55	**Letter + execution**	G10–G11 / S7–S8; sign, QR verify, bank acknowledgement.
5:55–7:00	**Settlement + audit**	G12; confirm AICTE share and verify audit chain.

Fallback: if an API/demo step fails, show the recorded seeded run and label it as a recording.

Source: SRS §9.1; 4-Day Plan §§1.1, 4

From a claim to a verifiable chain of evidence.

Submit → Confirm → Monitor → Release → Verify → Settle

Sakshya / Evidence before action.

Prototype v1.0 • SIH 2026 • Synthetic data • Integration status varies by module